

AI & Accessibility in Undergraduate Physics

A Resource Guide for Instructors

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Based on: *Artificial Intelligence in Physics Education: Opportunities for Students with Disabilities*

Introduction

Students with disabilities — including those with dyslexia, dyscalculia, ADHD, and Autism Spectrum Disorder — face a range of structural, cognitive, and social barriers in undergraduate physics courses. Many of these barriers stem not from students' abilities, but from instructional designs that assume a standardized mode of learning. At the same time, recent advances in artificial intelligence offer new tools to make physics education more flexible and accessible.

This resource guide is intended to help physics instructors take practical steps toward more inclusive teaching, drawing on evidence from recent research in physics education, special education, and educational technology. It is organized around three themes: understanding the barriers students face, integrating AI tools responsibly, and adopting Universal Design for Learning (UDL) principles.

Part 1: Understanding the Barriers Your Students Face

Before introducing new tools or technologies, it helps to understand the landscape of challenges that students with disabilities commonly encounter in physics courses.

Structural Barriers

These arise from the design of physics curricula themselves. High-stakes assessments, dense technical readings, and rapid conceptual progression can disproportionately disadvantage students who need additional processing time, alternative formats, or modified pacing. Research suggests that lower academic performance among disabled students often reflects *misaligned instructional design* rather than differences in ability or potential.

Cognitive Barriers

Cognitive barriers are particularly pronounced in physics, where students must simultaneously process linguistic, mathematical, and conceptual information. Students with dyslexia may struggle with reading-heavy problem statements; those with dyscalculia may face difficulties manipulating numerical relationships; and students with ADHD or ASD may find it difficult to sustain attention through extended derivations or abstract reasoning tasks.

Social Barriers

Social barriers are often overlooked but are quantitatively significant.

By the numbers:

33.7% of students with disabilities report experiencing exclusionary, intimidating, or hostile campus experiences, compared to **17.1%** of non-disabled students (study of 51,000+ students).

23% of students with disabilities had witnessed discrimination and **22%** had experienced offensive verbal comments — rates second only to those reported by African American students (study of 13,844 undergraduates).

Source: Harbour & Greenberg (2017), NCCSD Research Brief.

These experiences contribute to reluctance to self-advocate, seek accommodations, or ask for help — particularly in competitive STEM environments.

A Note on Self-Advocacy

Unlike secondary school, where accommodations are often coordinated by staff, post-secondary students are responsible for identifying themselves and requesting support — often course-by-course. This process is emotionally and cognitively taxing, and many physics instructors receive little training in inclusive pedagogy. Proactively creating a welcoming environment reduces this burden significantly.

Part 2: Integrating AI Tools Responsibly

AI tools are already being used by students in your courses, whether or not they are formally incorporated into your teaching. The question is not whether AI will be present, but how to integrate it in a way that supports learning rather than replacing it.

How Students Are Using AI

Students primarily use AI as a conceptual assistant — translating problem descriptions into equations, generating diagrams, clarifying derivations, and requesting alternative explanations. For students with disabilities in particular, AI offers something that traditional supports often do not: private, on-demand, judgment-free access to help. Students can ask the same question multiple times, explore different explanations, and engage with material at their own pace without fear of social stigma.

How Instructors Can Use AI

AI tools can assist you in several concrete ways:

- **Generate multiple explanations** of the same concept, making it easier to offer alternative representations for students with different cognitive profiles.
- **Produce visualizations and simulations** of abstract phenomena — such as wavefunction evolution, orbital motion, or electromagnetic fields — that would be time-consuming to build manually.

- **Provide automated feedback** on routine assignments, freeing you to focus on higher-order conceptual mentoring.
- **Analyze student performance data** to identify concepts that consistently challenge learners and guide course redesign.

Guidelines for Responsible Integration

Scaffold, don't replace. AI is most effective when it provides hints, partial explanations, or visual supports that guide students toward solutions — not when it produces complete answers. Design activities that require students to engage critically with AI outputs, such as identifying errors in AI-generated derivations or comparing multiple AI-provided solutions.

Build AI literacy. Students benefit from understanding how AI generates responses, including its limitations and tendencies toward confident-sounding errors. Incorporating explicit discussions of AI reliability, hallucination, and bias into your course helps students use these tools more critically.

Be aware of AI detection tools. Automated plagiarism detection systems frequently produce false positives for students with disabilities, particularly those using assistive writing technologies, text-to-speech tools, or structured composition aids. Avoid relying solely on automated detection; apply human judgment and context before acting on any flagged submission.

Establish clear, explicit guidelines. Clearly communicate what AI use is and is not permitted in your course, and distinguish between AI as a learning support and AI as a shortcut. Policies that are vague or overly restrictive may inadvertently penalize students who rely on AI as an accessibility tool.

Part 3: Universal Design for Learning (UDL) in Physics

Universal Design for Learning is a research-based instructional framework that encourages instructors to build flexibility into course design from the outset, rather than retrofitting accommodations after the fact. It is organized around three core principles:

1. **Multiple Means of Representation** — Present information in varied formats, including text, audio, diagrams, simulations, and worked examples. In physics, this might mean pairing mathematical derivations with visual animations, or offering both written and verbal explanations of conceptual material.
2. **Multiple Means of Action and Expression** — Offer students diverse ways to demonstrate their knowledge. This could include oral explanations, written reports, annotated diagrams, or problem-solving portfolios alongside traditional exams.
3. **Multiple Means of Engagement** — Provide varied opportunities for motivation and self-regulation. Low-stakes practice opportunities, flexible deadlines on formative work, and choice in how students engage with difficult concepts can all support sustained effort across diverse learners.

Research consistently shows that incorporating UDL principles into STEM course design improves accessibility without reducing disciplinary rigour. Importantly, these design choices benefit *all* students — not only those with identified disabilities.

Practical UDL Strategies for Physics

Strategy	Focus Area	Implementation Example
Multiple Explanation Formats	Reading Comp.	Written notes paired with LLM-generated summaries and simplified text.
Interactive Simulations	Physics Logic	Use AI to build or interpret field/wave visualizations (e.g., PhET).
Step-by-Step Scaffolding	Math Operations	AI tutors decomposing multi-step derivations into manageable logic blocks.
Low-Stakes AI Feedback	Engagement	Automated, non-judgmental support that allows for private trial-and-error.

Quick-Reference Recommendations

These recommendations are pulled directly from the formal study to guide the transition toward more accessible undergraduate physics environments.

- 1. Inclusive Design.** Adopt Universal Design for Learning (UDL) as the default framework for physics instruction. By providing multiple means of representation and expression, instructors reduce the self-advocacy burden on students with disabilities.
- 2. Evaluation of Bias.** Instructors must critically evaluate AI tools for algorithmic bias and accessibility before implementation. Tools should be vetted to ensure they do not underserve neurodivergent users or reinforce existing educational inequities.
- 3. Student Interaction.** Focus on building AI literacy rather than relying on detection. Students should be taught to recognize the limitations of AI—including "hallucinated" physics logic—enabling them to use these tools as skeptical learning assistants.
- 4. AI Utilization.** Formalize AI as an assistive technology within course policies. Distinguish between academic misconduct and the legitimate use of AI to support reading comprehension, math scaffolding, and cognitive load management.

Further Reading and Resources

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